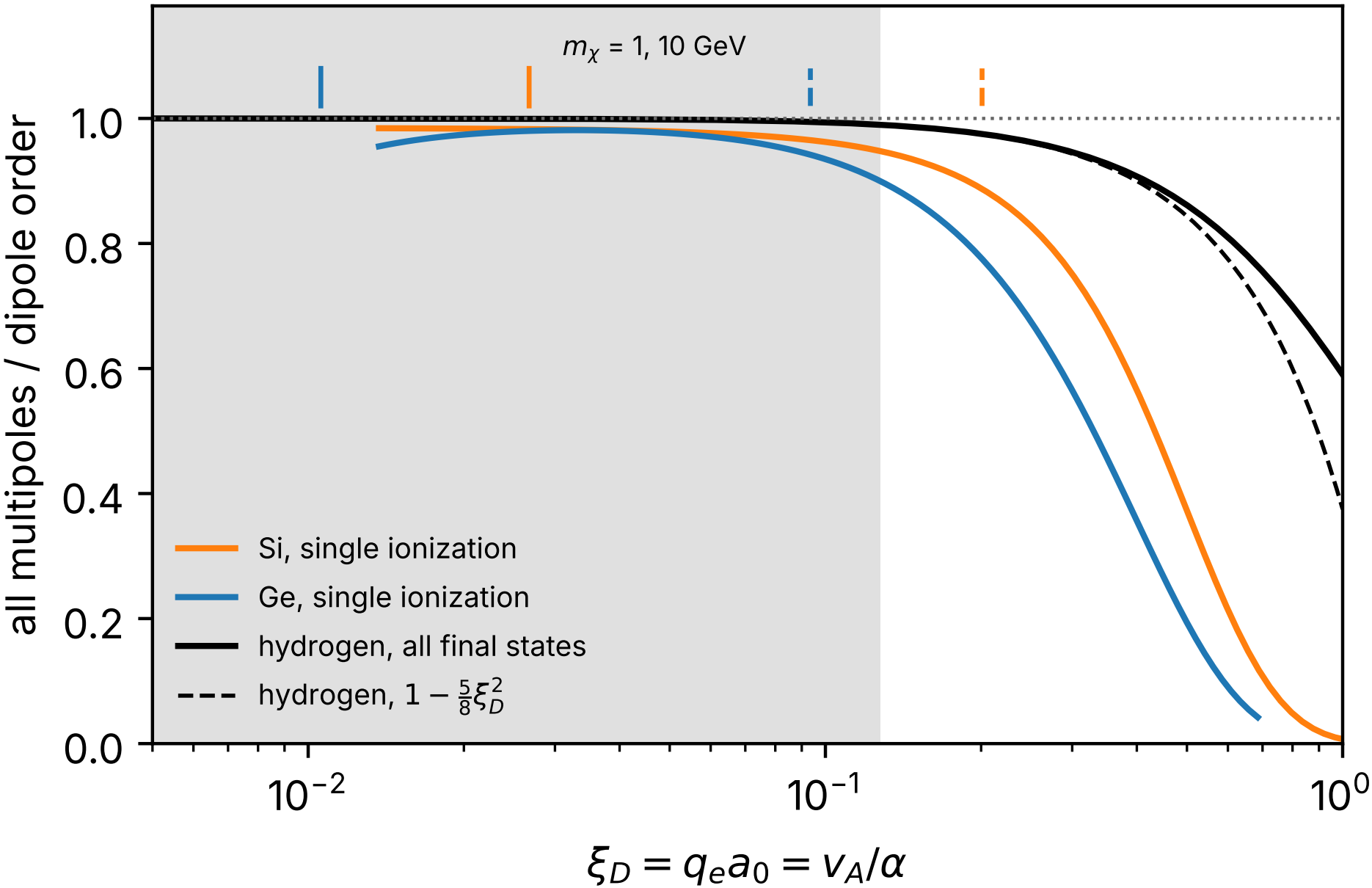
m_χ = 1, 10 GeV
all multipoles / dipole order
1.0
0.8
0.6
0.4
0.2
0.0
10^{-2}
10^{-1}
10^{0}
$\xi_D = q_e a_0 = v_A/\alpha$
Si, single ionization
Ge, single ionization
hydrogen, all final states
hydrogen, $1 - \frac{5}{8}\xi_D^2$